**Candidates' Name …….**

**Signature: ……...……**

| Random No. | | | | | Personal No. | | |
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***(Do not write your school / Center name or Number anywhere on this booklet***

***P525/1***
***CHEMISTRY***
***JUL / AUG, 2026***
**$2\frac{3}{4}$ Hours**

KAMPALA TEACHERS EDUCATION CONSULTANCY

**KAMTEC EXAMINATIONS BOARD**

**Uganda Advanced Certificate of Education**
**CHEMISTRY**
**PAPER 1**
**TIME: 2HOURS 45 MINUTES**

**INSTRUCTIONS TO CANDIDATES:**

- *The paper consists of section **A** and **B**.*
- *Section B consists of two parts **I** and **II***
- *Answer all items in section **A** and **two** items from section **B.***
- *Respond to **one** from each part.*
- *Illustrate your answers with equations where applicable.*

***TURN OVER***

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## SECTION A

(*Attempt **all** items in this section*)

***Item one***.

The government is encouraging the use of clean energy technologies as part of its efforts to reduce greenhouse gas emissions. A renewable-energy company is currently investigating the production of hydrogen fuel for use in fuel-cell-powered charging stations. Hydrogen is considered a promising alternative fuel because it can generate electricity without producing carbon dioxide.

Before constructing a pilot production plant, the company is evaluating a reaction that produces hydrogen gas from two aqueous reactants, **X** and **Y.**

X and Y react according to the following equation:

$$2X(aq) + Y(aq) \longrightarrow H_2(g) + WZ(aq)$$

A series of experiments conducted on the reaction above revealed the result summarized in Table 1 and Table 2 below:

**Table 1: Initial rate data**

| Experiment | [X] $(mol\ dm^{-3})$ | [Y] $(mol\ dm^{-3})$ | Initial rate $(mol\ dm^{-3}s^{-1})$ |
|---|---|---|---|
| 1 | 0.10 | 0.10 | $2.0 \times 10^{-3}$ |
| 2 | 0.20 | 0.10 | $8.0 \times 10^{-3}$ |
| 3 | 0.30 | 0.10 | $1.8 \times 10^{-2}$ |
| 4 | 0.10 | 0.20 | $4.0 \times 10^{-3}$ |

**Table 2: Thermochemical data**

| **Species** | $\Delta H_f^{\theta}(kJ\ mol^{-1})$ |
|---|---|
| X(aq) | $-120$ |
| Y(aq) | $-250$ |
| $WZ(aq)$ | $-670$ |

The company is currently unsure how this interpret the data and proceed with their plans. They have been advised that a rate constant above $2.0 \times 10^{-2}\ mol\ dm^{-3}s^{-2}$ and an enthalpy of reaction more negative than $-150\ kJ\ mol^{-1}$ represent an efficient production process.

Furthermore, the company lacks knowledge on how to increase the rate of hydrogen production. They intend to establish production units in remote communities where efficiency, safety, and sustainability are critical.

You and your teacher recently visited the company, and they provided you with details regarding their reactions and future plans.

**Task**

(a) Analyse the data provided and use it to comment on whether the reaction can actually be used for electricity generation and help the company assess the efficiency of the production process.

(b) Explain the conditions the company should optimise to increase the production rate of the fuel.

(c) Analyse the importance of these laws in the scenario for real-life applications and make a recommendation regarding the qualities of a good fuel that the company should consider.

**Item 2**

Acids and salts have a wide range of applications various fields, such as agriculture; however, care must be taken to avoid applying them in excess.

The "Valley Greens" community farm specializes in growing premium spinach, which requires a specific soil pH of 7.5 to 8.0 to thrive. For years, the farmers have used ammonium chloride as a nitrogen source to boast growth.

Recently, the plants have become stunted and yellow. A local soil expert was consulted and suspects that the excess ammonium chloride used is the likely cause. Laboratory analysis of the soil samples—to which ammonium chloride had been added—was conducted to determine the concentration of hydrogen ions. The results are summarized in **Table 3**:

**Table 3:**

| |
|---|
| • *20.0 g of the soil sample, to which ammonium chloride had been applied, was dissolved in 400 $cm^3$ of distilled water.* |
| • *20.0 $cm^3$ of the resulting filtrate required 5.0 $cm^3$ of 0.02 M sodium hydroxide solution for a complete reaction using methyl orange as an indicator.* |

Unfortunately, the expert could not provide further details following the laboratory analysis due to an urgent call she received.

Additionally, the farm manager needs to prepare 20 litres of a specific mixture using ammonia solution and ammonium chloride to maintain the soil at a target pH of 9.25 during a delicate seeding stage. He has about 200 g of ammonium chloride available and sufficient ammonia solution in his store with a concentration of 0.02M. He wants to ascertain whether the available ammonium chloride will be sufficient for his needs, but he doesn't know how to proceed.

($K_b$ for ammonia is $1.8 \times 10^{-5}\ mol\ dm^{-3}$ at 25℃)

You have been contacted by the farm manager for help.

**Task:**

(a) Use the data provided in **Table 3** to help the farmers justify why the spinach crops are failing in the soil.

(b) Help the farm manager

(i) ascertain if the available ammonium chloride is sufficient for the mixture he intends to prepare.

(ii) understand the how temperature fluctuations can affect his mixture.

(c) Explain other real-life applications of the mixtures that behave in a similar way to the mixture of ammonium chloride and ammonia.

## (SECTION B)

**(*Attempt two items; one from each part*)**

## PART I

***Item 3***

Water treatment and waste management systems are key in improving public health and environmental sustainability. A rapidly growing municipality in Uganda relies on chlorine for disinfecting drinking water, but recent expansion plans have raised concerns about the safe handling, storage, and disposal of chemicals used in water treatment and nearby industries. Several industries operating within the municipality use or produce substances containing chlorine. These include

- Sodium chloride (NaCl)
- Aluminium chloride ($AlCl_3$)
- Tin(IV) chloride ($SnCl_4$)
- Chlorine ($Cl_2$)

During stakeholder meetings, engineers, environmental officers, and public health officials observed that these substances behave very differently during storage, transport, and use. For example, some exist as solids with very high melting, while others are gases or volatile liquids. Some react readily with water to produce solutions that corroded the metal structures with the plant, while others dissolve with little or no chemical change. Similarly, when treated with sodium hydroxide during waste treatment, they undergo entirely different reactions. The stakeholders failed to understand these differences in behaviour.

To make informed decisions about chemical selection, handling, and disposal, the municipality needs to understand the above behaviors of the compounds but lack the skills and knowledge. This understanding would be used to develop guidelines for chemical use in water treatment, industrial processes, and environmental management.

The stakeholders have invited a nearly by school of which you are among the learners.

**Task.**

(a) Help the municipality to analyse and understand the behaviors of the compounds.
(b) Assess the use of chlorine by the municipality.

***Item 4***

Various oxides are used in industries to manufacture a wide range of products, such as cement, fertilizers, and ceramics. However, their properties significantly impact heating requirements, worker safety, and environmental pollution, particularly during industrial processing.

Farmers living near a certain factory recently reported crop damage and accelerated corrosion of iron sheets. Simultaneously, workers and engineers raised concerns regarding irritating oxide fumes and the high temperatures required to process certain oxide materials. In response to these concerns, the District Science and Environment Committee organised a joint industrial inspection involving chemists, environmental officers, and Senior Six chemistry learners from a neighboring school.

One of the industries visited utilizes oxides formed by Period 3 elements. During the inspection and testing, the learners observed that some oxides reacted vigorously with water to form corrosive solutions, while others dissolved only in alkaline solutions. The learners are keen to understand the reasons behind these differences.

The District Committee has requested that the learners prepare a scientific report to help industries improve safety, reduce pollution, and select appropriate oxide materials for industrial use. The data in Table 4 below outlines one of the properties of the selected oxides currently used in the factories visited.

**Table 4**

| Oxide | $Na_2O$ | MgO | $Al_2O_3$ | $SiO_2$ | $SO_3$ |
|---|---|---|---|---|---|
| Melting point (°C) | 1275 | 2827 | 2007 | 1607 | 30 |

**Task**:

As one of the Senior Six learners:

(a) Explain why some oxides produce fumes while others require very high temperatures to process.
(b) Analyse and explain the observed reactions of the oxides during inspection and testing.
(c) Assess the environmental, health, or industrial implications associated with the production and use of these oxides, and recommend suitable safety measures.

## PART II

***Item 5***

Organic compounds have a wide range of uses and can be converted into other useful substances through different chemical reactions. Recently, employees at a newly established chemical manufacturing company, which uses propan-1-ol and ethyne as raw materials to produce customer-required products, observed that these two compounds behave differently when heated. They do not understand why. The quality assurance team advised the staff that understanding the behaviour of organic compounds when exposed to heat is important, as it helps prevent potential explosions and other associated hazards.

A group of chemistry students who visited the factory during a science innovation exhibition became interested in how these simple organic compounds could be converted into more useful industrial products. During discussions with the industrial chemists, the students learnt that the company was particularly interested in improving the conversion of propan-1-ol into ethanol, and ethyne into 1-phenylethanol, because these products are needed in the manufacture of other goods. The students wondered how these transformations can be achieved.

To expand their production line, the company's director recently imported two additional raw materials: ethanal and benzene. The company has been advised to convert these into the more profitable products, ethanal 2,4-dintrophenylhydrazone and benzene sulphonic acid, respectively. This is because these two compounds are in high demand in the neighbouring industries as important precursors. However, the company's manager needs to understand how these conversions can be effectively carried out to minimize any potential dangers associated with the processes.

The students, company workers, and the company director have approached you for assistance.

Task:

(a) Prepare a write-up that will assist the learners, the company's workers, and the company's director.
(b) Evaluate the industrial importance and safety concerns associated with the raw materials in the above industrial processes.

***Item six.***
A cooperative of young entrepreneurs in Uganda plans to establish a small agroprocessing and chemical manufacturing enterprise. The enterprise intends to produce and market a range of products, including: a fruit-flavored beverage made from locally grown pineapples and mangoes, a fragrance additive for soaps and cosmetics, a cleaning solvent for industrial and laboratory use, and a biofuel additive to improve fuel combustion efficiency.
Before production begins, the entrepreneurs must select suitable organic compounds and production methods. The compounds considered for use are:

- Ethene ($C_2H_4$)
- Ethanol ($C_2H_5OH$)
- Ethanoic acid ($CH_3COOH$)
- Propan-1-ol ($C_3H_7OH$)
- Ethyl ethanoate ($CH_3COOC_2H_5$)

During preliminary trials, they observe that some compounds evaporate quickly, while others remain liquid for long periods; they are unsure why this occurs.

At the enterprise, benzene is reacted with a warm mixture of concentrated nitric acid and concentrated sulphuric acid, while propan-1-ol is heated with ethanoic acid in presence of sulphuric acid. The workers are eager to understand how these reactions proceed to yield the desired products.

Workers at the enterprise also discovered that several products can be made from the same starting materials. As a result, they are considering preparing propyl benzoate and an azo-dye from ethene, but they lack the necessary expertise.

Furthermore, the workers at enterprise lack an understanding of the which organic compounds are most suitable for the specific applications required in the production of these various goods.

**Task:**

(a) Explain the differences in the evaporation behaviour of the compounds considered for use.
(b) Help the workers in the cooperative solve their other challenges.
(c) Assess the suitability of one of these compounds for use in the production of a specified product the company intends to make.

***END***